

BARCELONA AP, Spain

WMO: 081810

Lat: 41.2928N

Lon: 2.0700E

Elev: 4

StdP: 101.28

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.9	3.1	-9.1	1.7	6.5	-5.9	2.3	6.6	11.6	12.3	10.2	11.7	5.3	350	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.0	30.9	23.5	29.8	23.4	28.9	23.1	25.4	29.2	24.6	28.5	23.9	27.8	4.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.0	28.2	23.2	18.0	27.6	22.6	17.3	27.1	77.5	29.2	74.5	28.5	71.7	27.9	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.2	8.8	7.8	-0.1	32.8	1.6	1.6	-1.3	33.9	-2.2	34.8	-3.1	35.7	-4.3	36.8		
			DB													
			WB	-2.4	26.1	1.9	1.3	-3.7	-4.9	27.9	-5.9	28.6	-7.3	29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.9	9.9	10.3	12.3	14.6	17.9	22.0	24.8	25.2	22.1	18.7	13.5	10.6
	DBStd	5.96	2.49	2.53	2.12	1.99	2.25	2.50	2.00	2.00	2.27	2.52	2.88	2.70
	HDD10.0	91	31	24	6	0	0	0	0	0	0	0	6	25
	HDD18.3	1234	260	226	186	112	35	2	0	0	2	26	146	241
	CDD10.0	2598	29	32	78	139	244	360	458	472	363	269	111	43
	CDD18.3	699	0	0	0	1	21	112	199	214	114	37	1	0
	CDH23.3	4776	0	0	0	1	41	590	1629	1840	597	78	1	0
	CDH26.7	1041	0	0	0	0	2	98	376	489	72	3	0	0
Wind	WSAvg	3.9	4.2	4.1	4.1	4.2	3.9	3.7	3.7	3.7	3.8	3.8	4.0	4.3
Precipitation	PrecAvg	623	42	34	40	48	49	42	22	58	80	100	62	56
	PrecMax	1052	208	153	125	174	232	107	94	177	277	411	291	254
	PrecMin	405	0	0	0	3	8	0	0	0	0	0	0	0
	PrecStd	163	43	35	31	31	41	29	28	40	67	85	66	56
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	19.2	20.9	22.6	26.2	30.9	31.8	33.5	29.9	27.0	22.1	18.9
		MCWB	13.2	12.2	12.4	14.4	17.7	21.6	23.3	24.2	22.3	20.4	17.2	13.5
	2%	DB	16.8	17.1	18.8	21.0	24.3	29.1	30.5	31.5	28.4	25.1	20.6	17.1
		MCWB	11.7	11.6	12.6	14.6	17.5	22.1	23.6	23.8	22.5	20.4	16.2	13.0
	5%	DB	15.2	15.7	17.2	19.7	23.0	27.7	29.5	30.1	27.3	24.1	19.3	16.0
		MCWB	11.2	11.2	12.4	14.5	17.2	21.2	23.4	23.7	22.0	19.7	15.3	12.5
	10%	DB	14.1	14.6	16.2	18.6	22.0	26.3	28.7	29.1	26.4	23.0	18.2	15.0
		MCWB	10.7	10.8	12.2	14.1	17.1	20.7	23.1	23.5	21.2	19.0	14.6	11.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.2	13.9	15.1	16.9	20.0	24.6	26.0	26.7	24.9	22.7	18.6	15.2
		MCDB	17.1	17.0	17.9	19.7	23.3	28.4	29.7	30.5	27.9	24.8	20.7	17.0
	2%	WB	13.2	12.9	14.2	16.0	19.1	23.5	25.1	25.7	23.7	21.5	17.5	14.2
		MCDB	15.6	15.5	16.8	19.0	22.4	27.3	29.0	29.6	27.1	24.0	19.8	16.3
	5%	WB	12.4	12.1	13.6	15.3	18.4	22.5	24.4	25.0	22.9	20.6	16.4	13.2
		MCDB	14.7	14.7	16.1	18.4	21.8	26.3	28.4	28.8	26.6	23.3	18.6	15.5
	10%	WB	11.3	11.4	12.9	14.7	17.6	21.5	23.7	24.2	22.1	19.6	15.3	12.2
		MCDB	13.7	14.0	15.4	17.8	21.0	25.4	27.8	28.1	25.7	22.4	17.6	14.6
Mean Daily Temperature Range	5% DB	MDBR	7.9	8.0	7.7	7.3	7.0	6.9	6.7	7.0	7.3	7.2	7.8	7.8
		MCDBR	8.6	9.0	8.9	8.7	8.1	7.8	7.3	7.7	7.7	7.8	8.3	8.2
		MCWBR	6.0	6.1	5.7	5.3	5.0	5.1	5.2	5.5	5.1	5.4	6.0	5.9
	5% WB	MCDBR	7.5	8.0	7.7	7.4	7.1	7.0	6.9	7.2	7.1	7.1	7.3	7.3
MCWBR		5.8	5.9	5.6	5.4	5.1	5.2	5.4	5.6	5.2	5.4	6.0	5.9	
Clear-Sky Solar Irradiance	taub	0.326	0.350	0.395	0.411	0.432	0.445	0.453	0.450	0.428	0.403	0.356	0.328	
	taud	2.507	2.439	2.316	2.290	2.273	2.273	2.272	2.284	2.335	2.385	2.467	2.504	
	Ebn at Noon	828	856	850	861	848	835	824	815	807	783	780	792	
	Edh at Noon	74	92	116	127	133	133	132	127	113	96	77	69	
All-Sky Solar Radiation	RadAvg	2.09	2.99	4.33	5.40	6.39	7.17	7.09	6.14	4.67	3.31	2.27	1.83	
	RadStd	0.17	0.31	0.32	0.38	0.44	0.27	0.31	0.29	0.23	0.27	0.20	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.35	N/A	-1.46	N/A	N/A	N/A	N/A	N/A	+117	+89
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+83	+59

Nomenclature: See separate page